ZCAS UNIVERSITY

BACHELOR OF SCIENCE IN COMPUTER SCIENECE

CIT34411 HUMAN COMPUTER INTERACTION

FINAL EXAMINATION

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

DATE : 17th JUNE, 2026

INSTRUCTIONS:

1. Section A is **40 marks**: this question is **compulsory** and must be attempted.
2. Sections B: Answer **any THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. **Candidates must not turn this page until the invigilator tells them to do so.**

SECTION A – COMPULSORY

QUESTION ONE

Read the scenario below and answer the questions that follow.

A large metropolitan hospital recently implemented an integrated digital healthcare management system designed to support patient registration, electronic medical records, appointment scheduling, laboratory reporting, prescription management, and communication between healthcare professionals. The system is accessible through desktop computers, tablets, and mobile devices to allow doctors, nurses, pharmacists, and administrators to perform tasks in real time. Although the institution invested heavily in the system to improve efficiency and quality of healthcare delivery, a post-implementation usability investigation revealed serious HumanComputer Interaction (HCI) challenges affecting both staff performance and patient safety.

Healthcare workers reported that the system interface was highly complex and cognitively demanding, especially during emergency situations where rapid decision-making was required. Doctors frequently struggled to locate critical patient information because of deeply nested menus, poor information hierarchy, and cluttered dashboards displaying excessive data simultaneously. Nurses complained that the system required repeated authentication and unnecessary navigation steps before administering medication or updating patient records. In several incidents, healthcare professionals accidentally selected incorrect patient profiles due to poor visual differentiation between records and inadequate confirmation mechanisms. The evaluation further revealed that inconsistent interaction styles across modules increased learning difficulties, particularly among newly recruited staff members who required extensive training before using the system confidently.

The investigation also identified major accessibility, interaction, and feedback-related issues. The system relied heavily on colour coding without alternative indicators, making it difficult for colourblind users to distinguish alerts and warnings. Some touch-screen controls were too small and closely positioned, resulting in frequent input errors on tablets used in intensive care units. Error messages were highly technical and failed to provide meaningful recovery guidance, often forcing staff to restart tasks entirely. Furthermore, system latency during peak operational hours caused delays in retrieving laboratory results and processing emergency prescriptions, increasing frustration and reducing trust in the system. Hospital management is therefore seeking assistance from HCI specialists to redesign the system using advanced usability engineering principles in order to improve safety, accessibility, efficiency, learnability, and overall user experience in critical healthcare environments.

(a) Critically examine the interdisciplinary nature of Human-Computer Interaction (HCI) by discussing how concepts from psychology, computer science, ergonomics, and cognitive science contribute to the design of interactive systems. **[8 marks]**

(b) Using appropriate HCI theories and usability heuristics, critically analyze the interaction design failures present in the healthcare management system. Your answer should demonstrate how these failures may compromise user performance, patient safety, and organizational productivity. **[12 marks]**

(c) Evaluate the impact of cognitive load, mental models, and human memory limitations on user interaction within the scenario. Support your discussion with practical examples from the case study. **[8 marks]**

(d) Apply suitable HCI models, frameworks, or design principles to recommend advanced interface and interaction improvements for the healthcare system. Your answer should include justification for each recommendation in relation to usability, accessibility, and error reduction. **[8 marks]**

(e) Critically discuss the role of usability testing, prototyping, and user-centred evaluation methods in the development of safety-critical systems such as healthcare applications. **[4 marks]**

[Total: 40 Marks]

SECTION B: Answer any three (3) questions in this section.

QUESTION TWO

Read the scenario below and answer the questions that follow.

A university has introduced an online learning system that allows students to access lecture notes, submit assignments, check results, and communicate with lecturers. The system is accessed using laptops, smartphones, and desktop computers. Some students find the system easy to use because they are familiar with technology, while others struggle because they have limited computer experience. The system uses different forms of interaction such as typing on a keyboard, clicking with a mouse, touching a screen, and sometimes speaking into a microphone. It also provides feedback through on-screen messages, notifications, and sound alerts to guide users during tasks.

The university aims to ensure that the system is easy to use, efficient, safe, accessible to all students, and satisfying to interact with.

(a) Define Human-Computer Interaction (HCI) and explain the three main components involved in HCI. **[6 marks]**

(b) Using examples from the scenario, explain how the user component in HCI affects system interaction. **[4 marks]**

(c) Explain any four goals of HCI and briefly state why each goal is important in the given online learning system. **[8 marks]**

(d) Identify two input methods and two output methods used in the system and explain their role in interaction. **[2 marks]** **[Total: 20 Marks]**

QUESTION THREE

(a) Explain the differences between Command-Line Interface (CLI) and Graphical User Interface (GUI), giving one example for each. **[6 marks]**

(b) Describe how Natural User Interfaces (NUI) are used in modern devices. Use two examples to support your answer. **[4 marks]**

(c) Explain the advantages of using Web Interfaces in services such as online banking or e-commerce platforms. **[6 marks]**

(d) Discuss Virtual and Augmented Reality interfaces and explain two real-life applications where they are used. **[4 marks]**

[Total: 20 Marks]

QUESTION FOUR

(a) Explain three emerging trends in Human-Computer Interaction (HCI) systems: self-healing systems, predictive disaster detection, and automatic recovery without human input. **[6 marks]**

(b) Describe how predictive systems can help businesses prevent disasters before they happen. Give suitable examples. **[4 marks]**

(c) Explain the role of IoT, cloud computing, and edge computing in enhancing future HCI systems. **[6 marks]**

(d) State and explain two advantages and two challenges of fully autonomous future HCI systems. **[4 marks]**

[Total: 20 Marks]

QUESTION FIVE (5)

(a) Define User-Centered Design (UCD) and outline its key characteristics. **[3 marks]**

(b) Describe the main stages involved in the User-Centered Design (UCD) process. **[7 marks]**

(c) Explain the importance of user feedback and usability testing in improving system design and performance. **[5 marks]**

(d) As a system designer, explain how you would apply User-Centered Design (UCD) when developing a student information system. **[5 marks]**

[Total: 20 Marks]